

Hayden Fuller

Modern Communication Systems F25

Lab 6

11/14/2025

1) Theory:

1. 1x2 MIMO

$$r = h^*_{A} y_A + h^*_{B} y_B$$

$$\hat{x} = \text{sign}(\text{real}(r))$$

SNR: $\gamma_A = |h_A|^2 E_s / \sigma_n^2$, B is similar

$$\gamma_{1x2} = (|h_A|^2 + |h_B|^2) E_s / \sigma_n^2$$

2. 1x4 MIMO

$$r = h^*_{A} y_A + h^*_{B} y_B + h^*_{C} y_C + h^*_{D} y_D$$

$$\hat{x} = \text{sign}(\text{real}(r))$$

$$\gamma_{1x4} = (|h_A|^2 + |h_B|^2 + |h_C|^2 + |h_D|^2) E_s / \sigma_n^2$$

this is higher than the first case with the addition of two more positive terms in the numerator, so all else equal this gives better SNR and lower BER

3. 2x1 MIMO

$$[c_A \ c_B] = \alpha [h^*_A \ h^*_B]$$

$$|\alpha|^2 (|h_A|^2 + |h_B|^2) = 1$$

$$\alpha = 1 / \sqrt{|h_A|^2 + |h_B|^2}$$

$$h_{\text{eff}} = h_{Ac} + h_{Bc} = \alpha (h_A^* h_A + h_B^* h_B) = \alpha (|h_A|^2 + |h_B|^2)$$

$$h_{\text{eff}} = \sqrt{|h_A|^2 + |h_B|^2}$$

$$y = h_{\text{eff}} x + n$$

$$\hat{x} = \text{sign}(\text{real}(y h_{\text{eff}}^*)) = \text{sign}(\text{real}(\sqrt{|h_A|^2 + |h_B|^2}))$$

$$\gamma_{2x1} = |h_{\text{eff}}|^2 E_s / \sigma_n^2 = (|h_A|^2 + |h_B|^2) E_s / \sigma_n^2$$

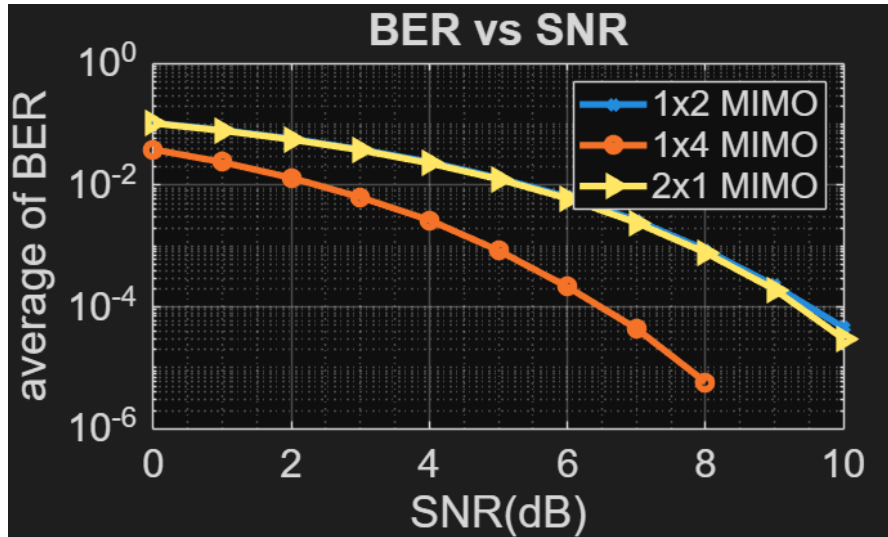
comparison:

between 2x1 and 1x2, they give the same combined SNR, MRT/MRC duality

1x4 is better than either because it sums four per branch gains

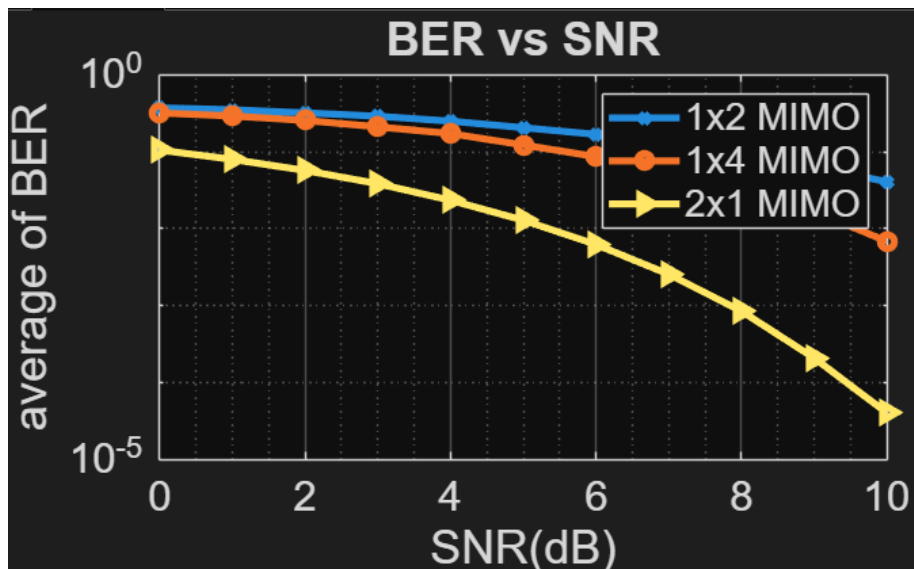
MATLAB:

2) default:



Yes, the BER values are as expected, the same for 1x2 and 2x1, and lower for 1x4, all decreasing with increased SNR. For the same BER target, 1x2 and 2x1 require a higher SNR than 1x4 since 1x4 sums 4 per branch gains.

3) USE_CHANNEL_ESTIMATES = 1;



In this version, 2x1 remained the same, while 1x2 and 1x4 increased BER significantly. These estimations create errors, and with more receiving antennas, there's a greater effect.

4) we would obtain the per-subcarrier channel estimates from LTS in packet_detection_and_channel_estimation, then for each OFDM subcarrier, form MRT/MRC precoding weights by taking the conjugate of each estimate and normalizing them to keep constant total transmit power